

5.2.1

Use the technique illustrated at the beginning of this section to show that the statements in (a) and (b) are true.

a.

$$\begin{aligned} &\text{If } \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) = \frac{1}{5} \\ &\text{then } \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{6}\right) = \frac{1}{6} \end{aligned}$$

b.

$$\begin{aligned} &\text{If } \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{6}\right) = \frac{1}{6} \\ &\text{then } \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{6}\right) \left(1 - \frac{1}{7}\right) = \frac{1}{7} \end{aligned}$$

a.

$$\begin{aligned} &\left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) = \\ &\left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{3}{4}\right) \left(\frac{4}{5}\right) = \frac{1 \times 2 \times 3 \times 4}{2 \times 3 \times 4 \times 5} = \frac{4!}{5!} = \frac{24}{120} = \boxed{\frac{1}{5}} \\ &\left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{6}\right) = \\ &\left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{3}{4}\right) \left(\frac{4}{5}\right) \left(\frac{5}{6}\right) = \frac{1 \times 2 \times 3 \times 4 \times 5}{2 \times 3 \times 4 \times 5 \times 6} = \frac{5!}{6!} = \frac{120}{720} = \boxed{\frac{1}{6}} \end{aligned}$$

Solution:

$$\begin{aligned} &\prod_{i=2}^5 \left(1 - \frac{1}{i}\right) = \frac{1}{5} \\ &\prod_{i=2}^6 \left(1 - \frac{1}{i}\right) = \frac{1}{6} \end{aligned}$$

b.

$$\begin{aligned} &\left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{6}\right) = \\ &\left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{3}{4}\right) \left(\frac{4}{5}\right) \left(\frac{5}{6}\right) = \frac{1 \times 2 \times 3 \times 4 \times 5}{2 \times 3 \times 4 \times 5 \times 6} = \frac{5!}{6!} = \frac{120}{720} = \boxed{\frac{1}{6}} \end{aligned}$$

$$\begin{aligned} &\left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{6}\right) \left(1 - \frac{1}{7}\right) = \\ &\left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{3}{4}\right) \left(\frac{4}{5}\right) \left(\frac{5}{6}\right) = \left(\frac{6}{7}\right) = \frac{1 \times 2 \times 3 \times 4 \times 5 \times 6}{2 \times 3 \times 4 \times 5 \times 6 \times 7} = \frac{6!}{7!} = \frac{720}{5,040} = \boxed{\frac{1}{7}} \end{aligned}$$

Solution:

$$\begin{aligned} \prod_{i=2}^6 1 - \frac{1}{i} &= \frac{1}{6} \\ \prod_{i=2}^7 1 - \frac{1}{i} &= \frac{1}{7} \end{aligned}$$